

NICE RIDE MINNESOTA 2017

DSB3
Done: WADEEA

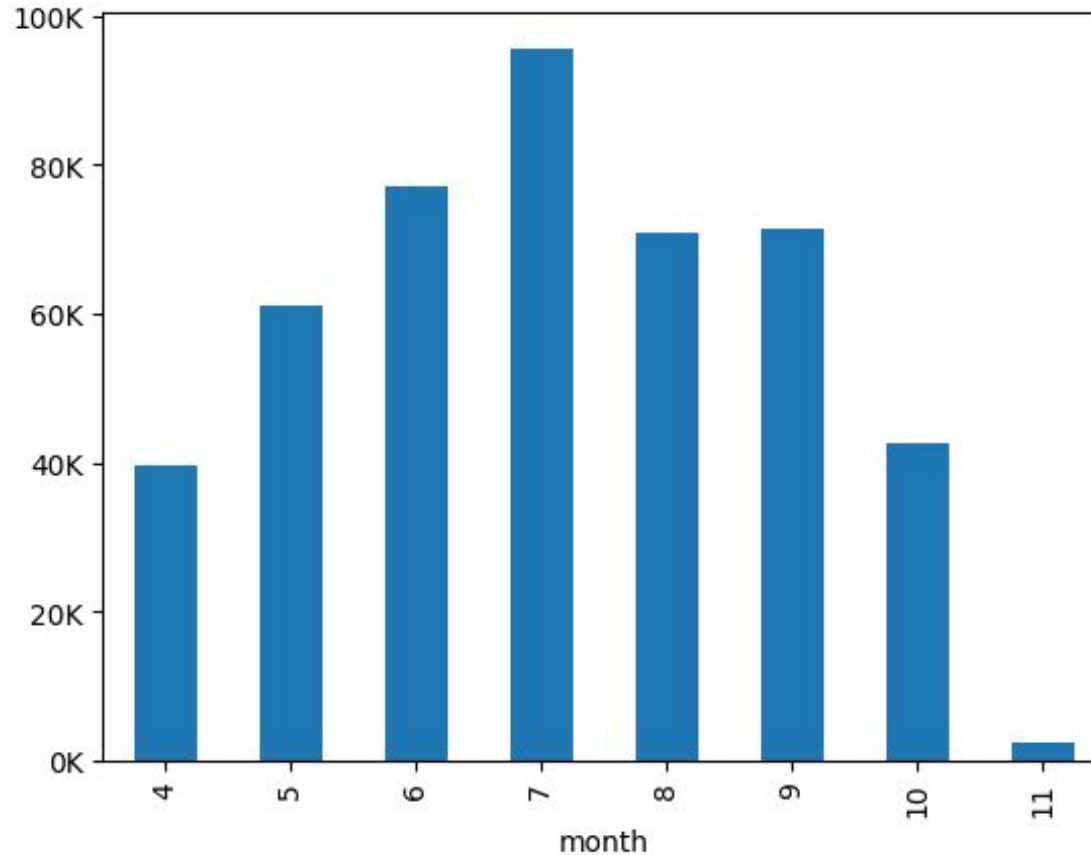
INTRODUCTION:

This project will analyze trip, station, and weather data to identify the periods and stations with the highest demand, compare demand with station capacity, and examine how weather affects bike usage

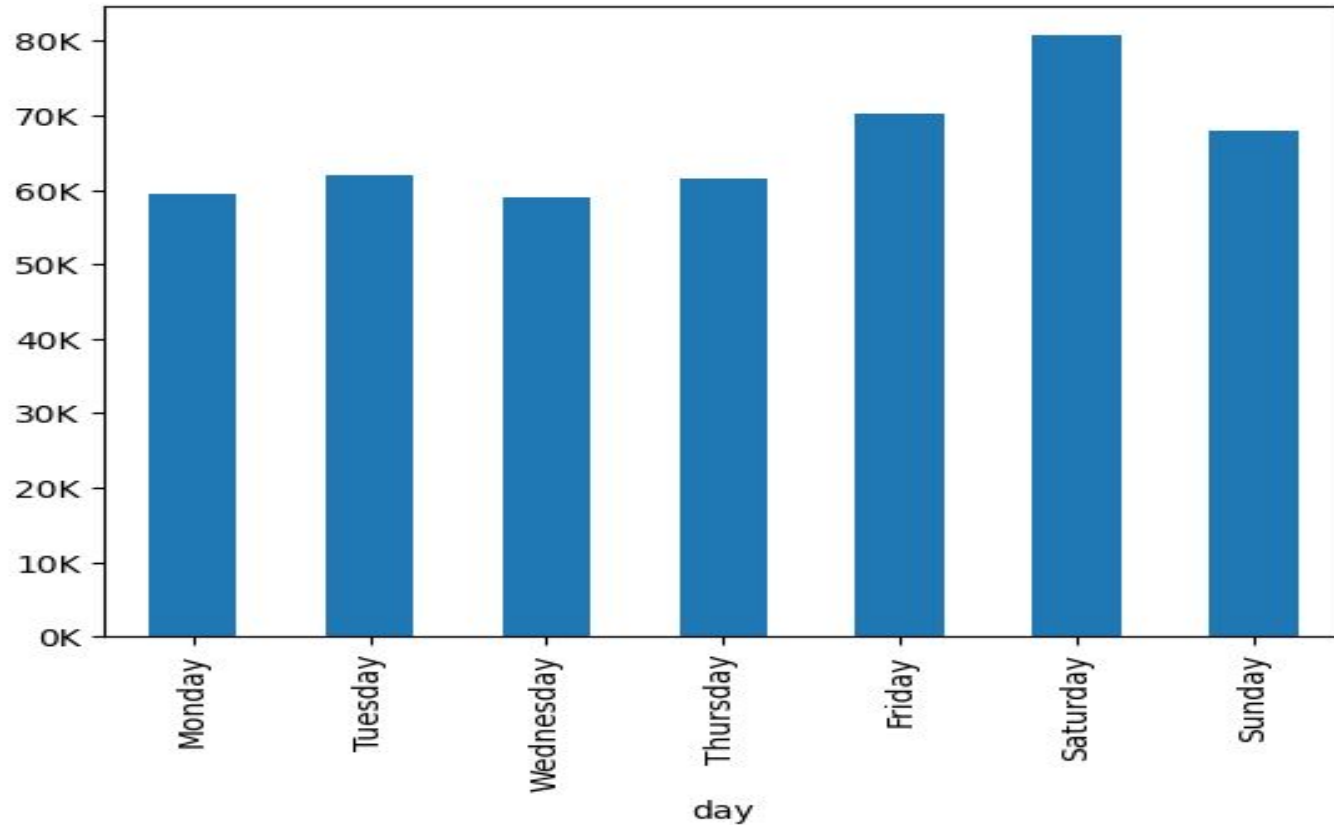
PROBLEM STATEMENT:

In **2017**, Nice Ride Minnesota recorded **460,718** bike trips across **202** stations. Bike usage varied across different **times**, **stations**, **user types**, **trip durations**, and **weather** conditions. Understanding these patterns can help identify periods of high demand, differences in user behavior.

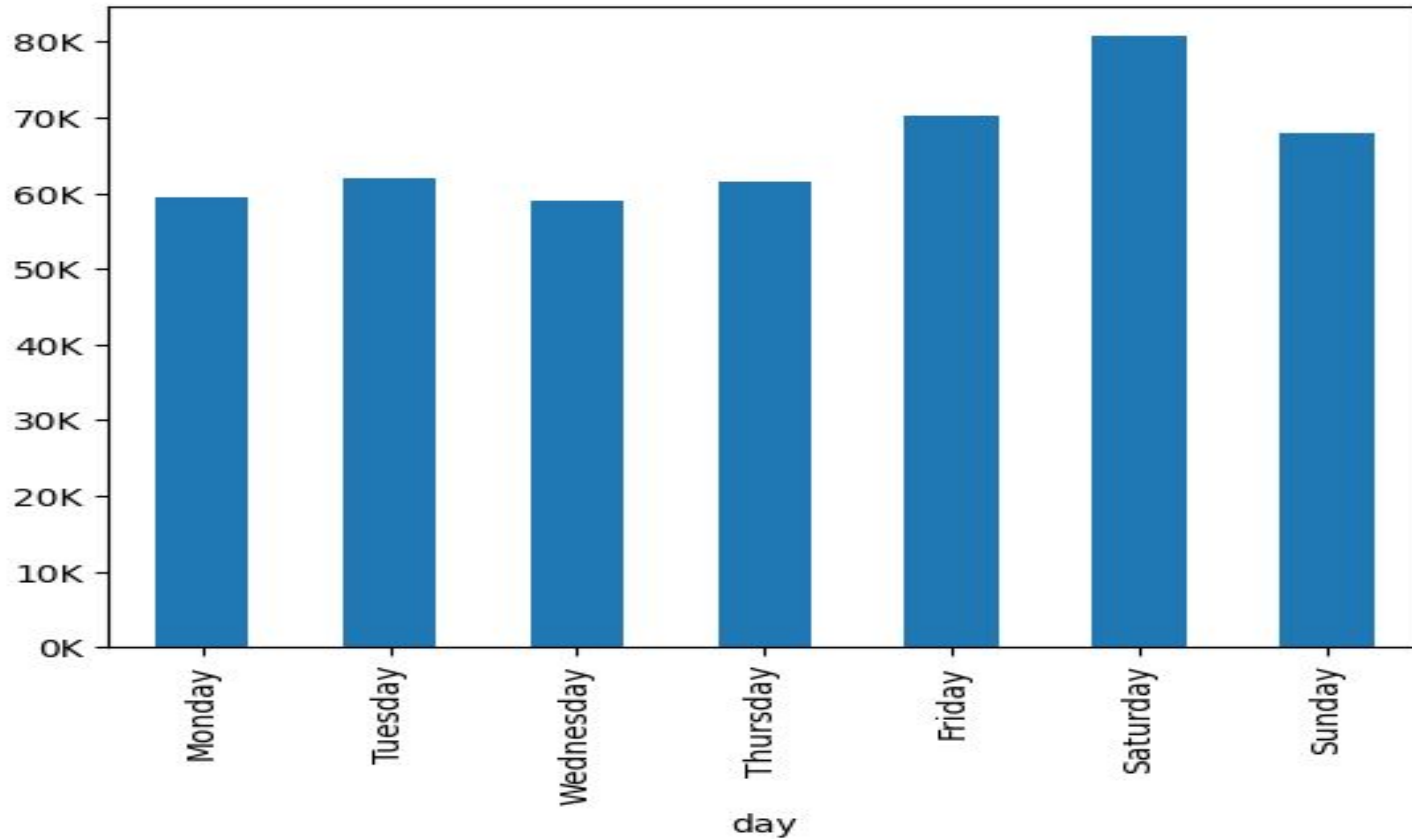
1-HOW DOES BIKE DEMAND CHANGE BY MONTH, ?



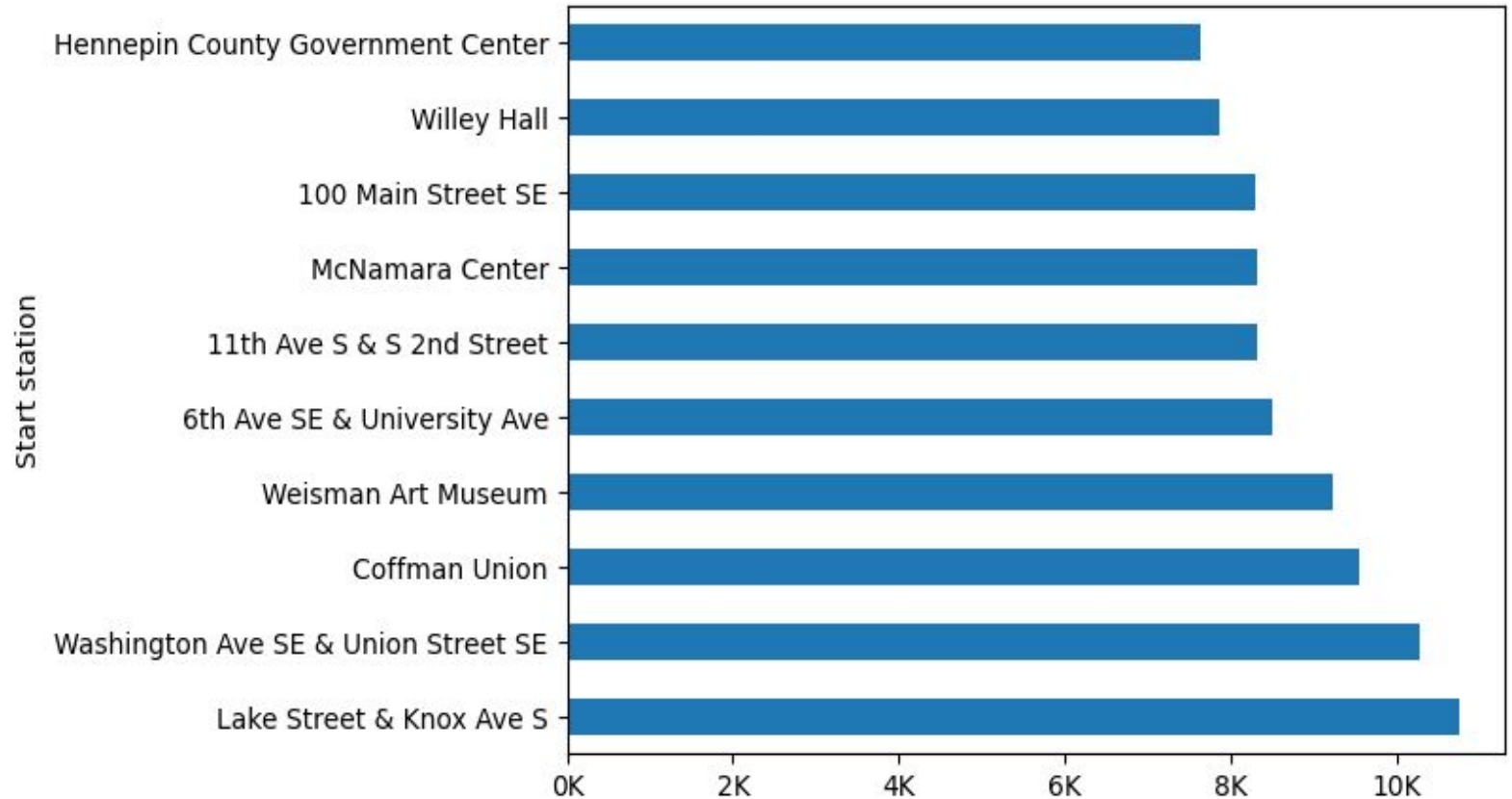
1-HOW DOES BIKE DEMAND CHANGE BY DAY ?



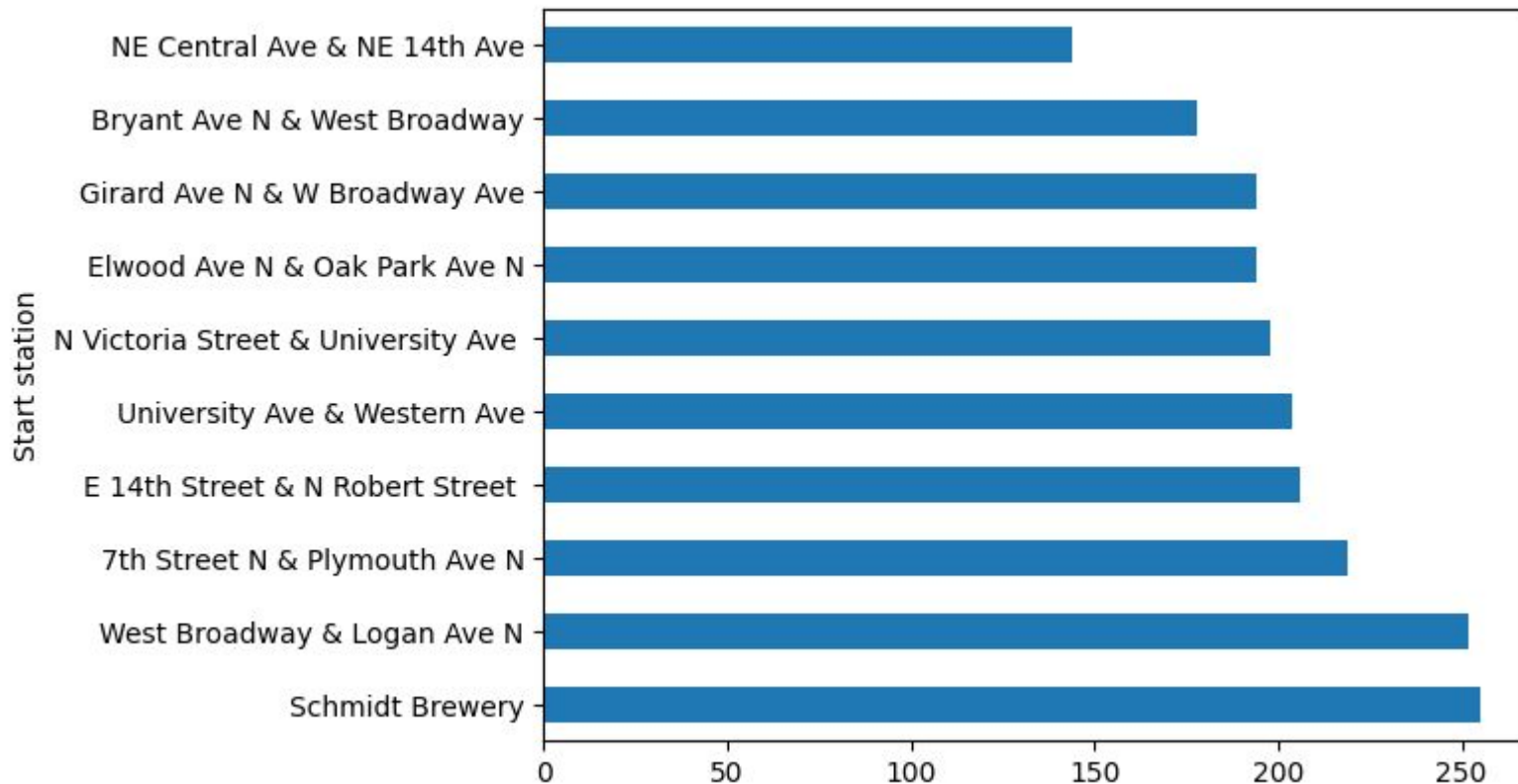
1-HOW DOES BIKE DEMAND CHANGE BY HOUR?



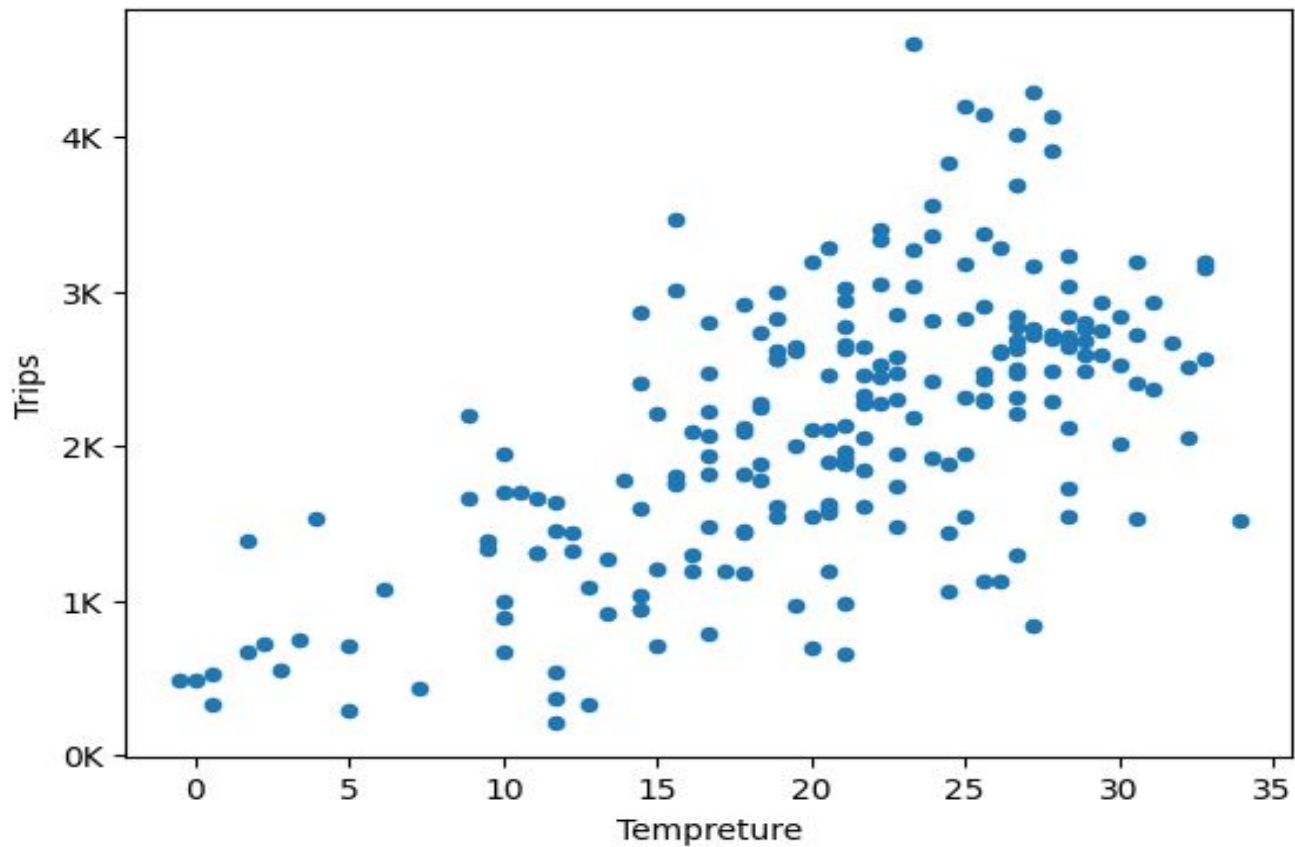
2-WHICH STATIONS HAVE THE HIGHEST BIKE DEMAND?



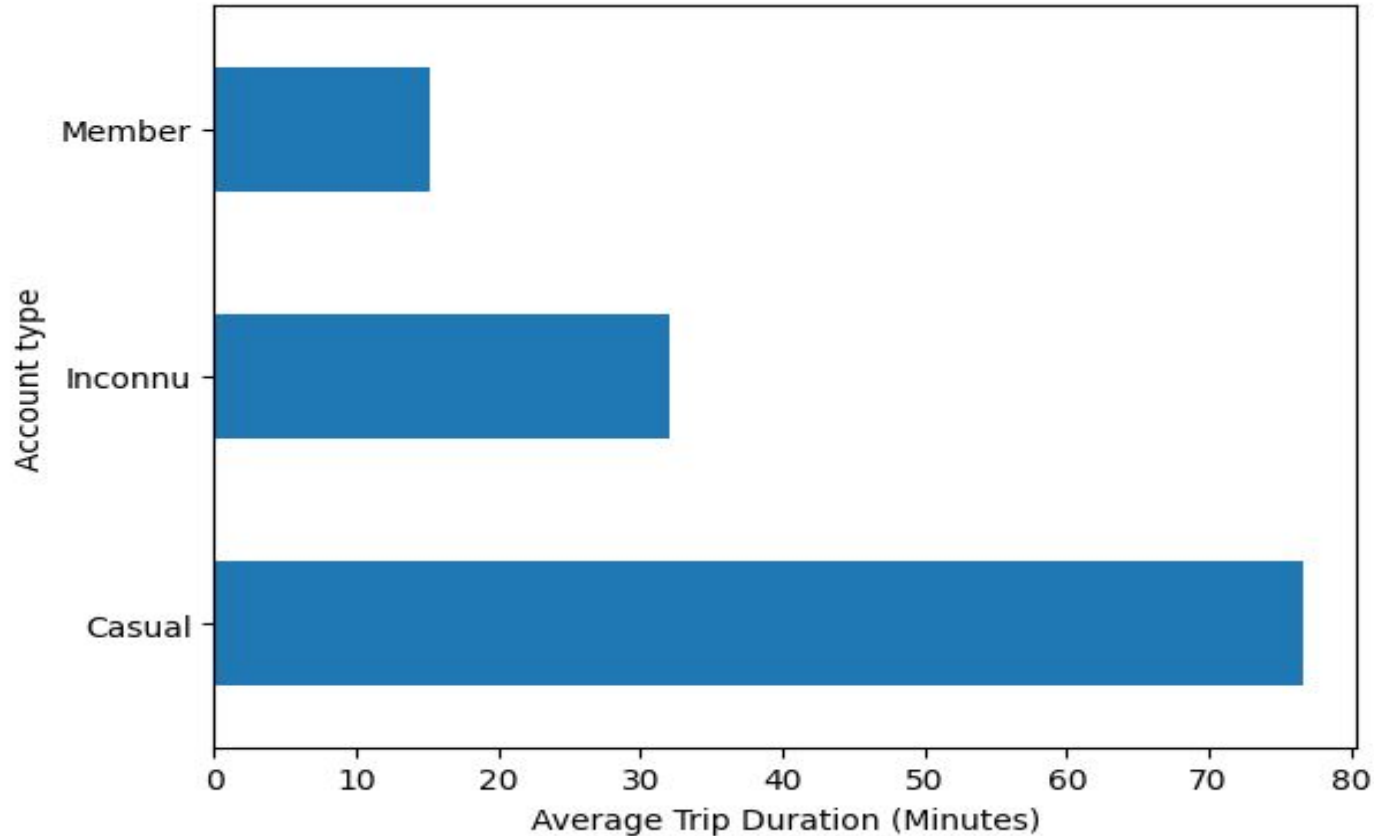
2-WHICH STATIONS HAVE THE LOWEST BIKE DEMAND?



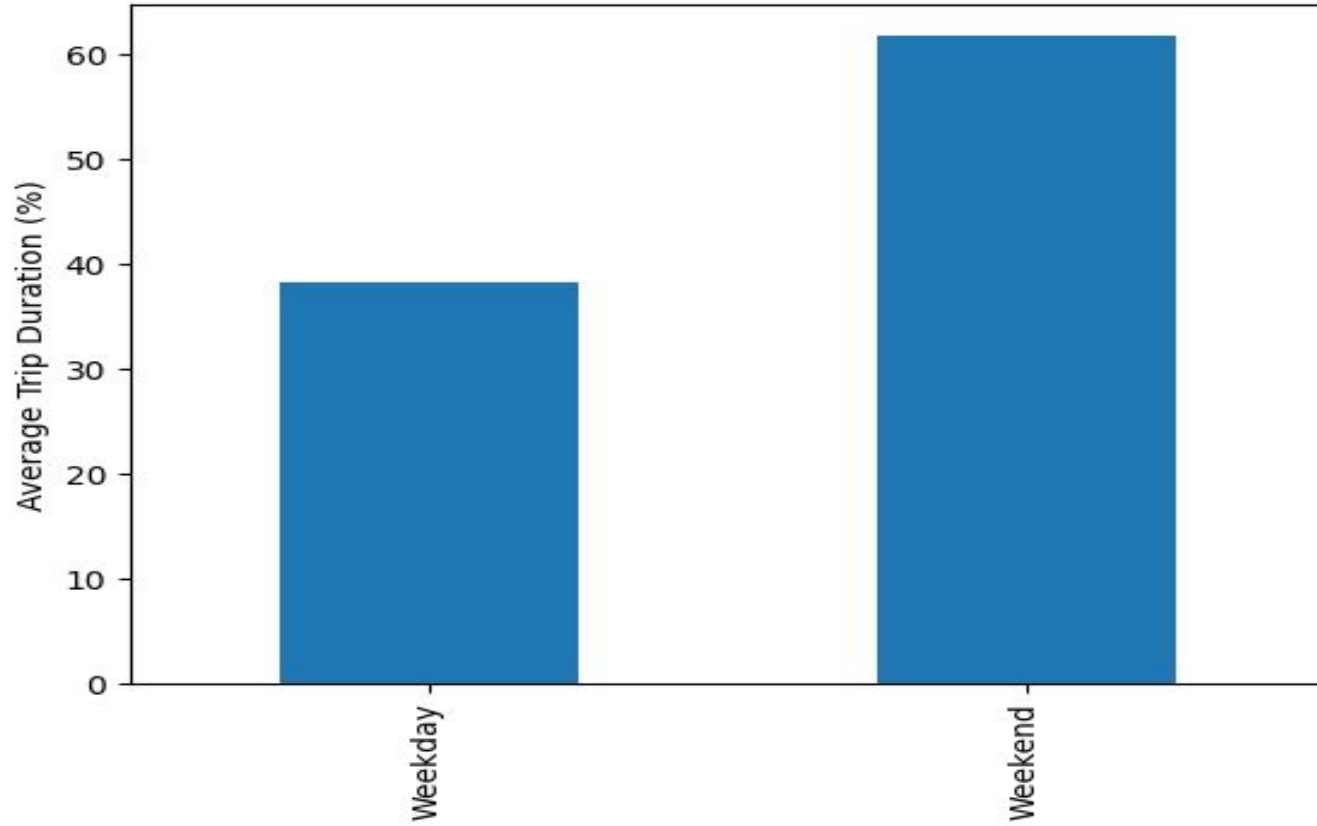
3-HOW DOES WEATHER AFFECT DAILY BIKE DEMAND?



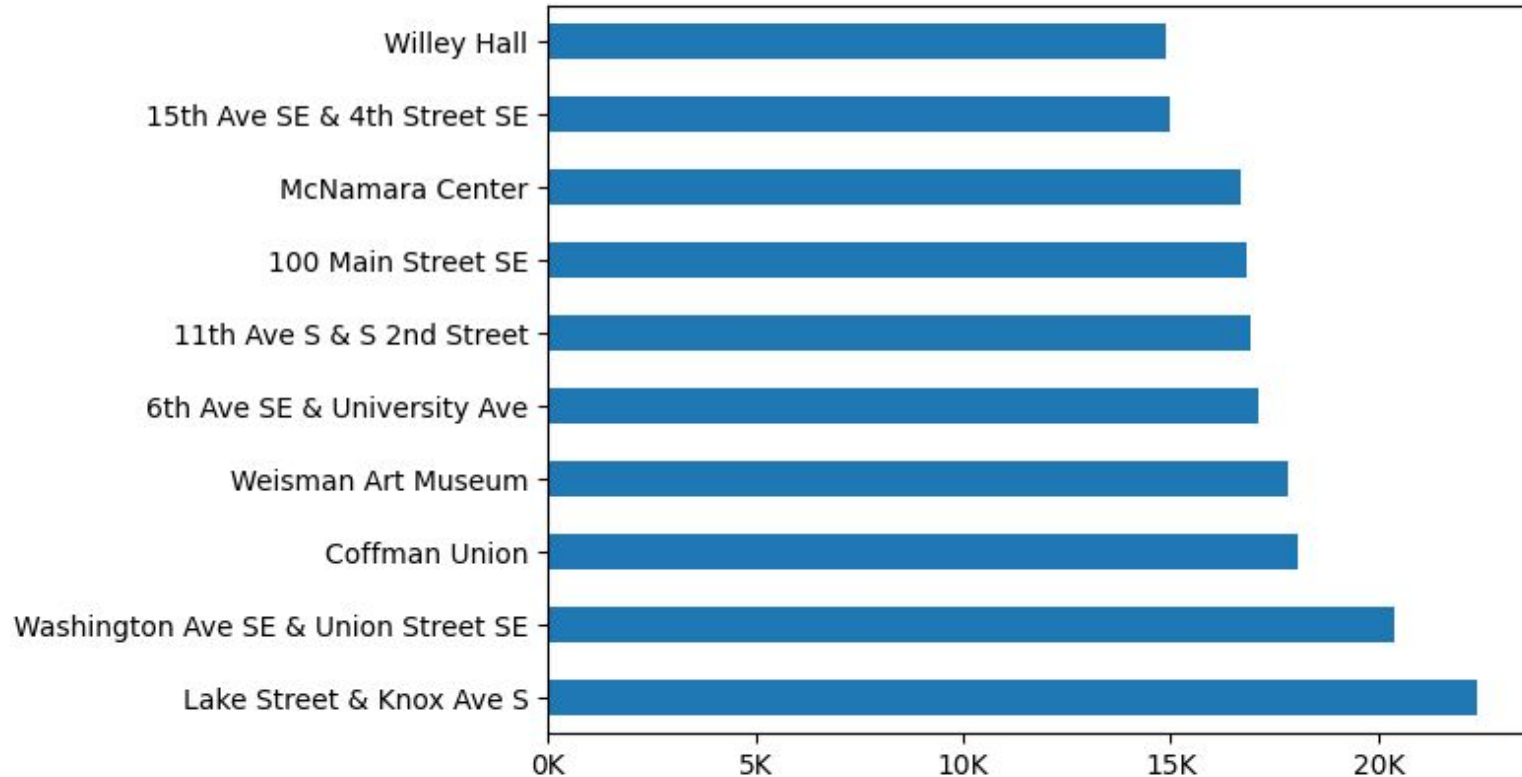
4- DO MEMBERS AND CASUAL USERS USE BIKES DIFFERENTLY?



5- ARE BIKE TRIPS LONGER ON WEEKENDS OR WEEKDAYS?



6- WHICH STATIONS ARE MOST COMMONLY USED AS BOTH START AND END STATIONS?



CONCLUSION :

- Bike demand showed clear variations across different times and stations, highlighting specific periods and locations with higher levels of activity.
- Weather conditions also influenced daily bike usage, suggesting that external factors can affect overall demand throughout the year.

RECOMMENDATION :

- Increase bike availability at high-demand stations.
- Add more bikes during peak hours.
- Improve bike distribution between stations.
- Consider weather when planning bike availability.
- Understand member and casual user patterns.

NEXT STEP:

- Further analysis could explore user types, trip duration, and station usage to identify different usage patterns.
- These insights could support better bike availability and more efficient station planning.